LAB TITLE:- INVESTIGATING DJI MAVIC AIR FLIGHT EVIDENCE

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STUDENT ID:- COMPUTER FORENSICS CASE STUDIES

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DATE OF SUBMISSION:- 2/11/2026

Case scenario

Investigators recovered an Android logical acquisition associated with the DJI GO 4 application and two representations of DJI Mavic Air internal microSD media. The enquiry is whether the sources support a reliable account of aircraft activity, recorded media, and the relationship between the mobile application and microSD evidence. The case material may contain flight locations, images, and records relating to people. This is an offline examination of authorised course artefacts. It does not authorise you to fly, power on, connect to, synchronise, reset, or change settings on any drone, mobile device, account, cloud service, or removable media.

Lab Objectives

The objectives of the lab is to understand how to analyse a digital evidence relating to a drone system.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- file:- determine file type
- ewfinfo:- is a utility to show meta data stored in EWF files
- tree:- list the contents inside a directories in a tree like format
- stat:- display file or file system status
- xxd:- make a hex dump or do the reverse
- lsblk :-list information about all the available or specific block device
- ewfvrify:- is a utility to verify media data stored in EWF files.
- ewfmount:- is a utility to mount data stored in EWF files
- umount:- this command is used for unmounting a filesystem
- icat:- output the contents of a file based on the inode numbers

- fls:- list the file and directory name in a disk image and can display the file name of recently deleted file for the directory using the given inode number

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Methology

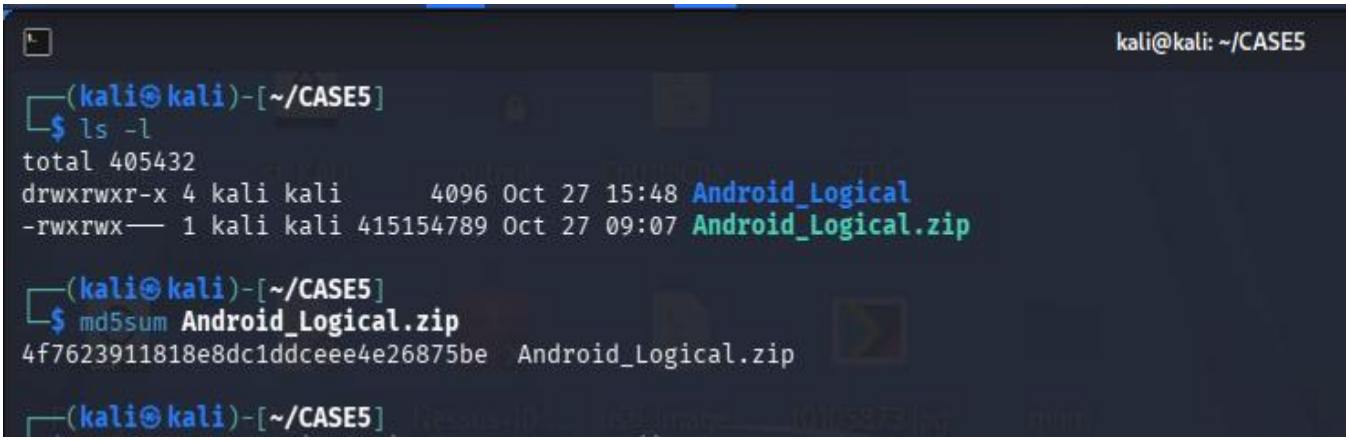
- I power on the kali linux machine
- I used the wget command to download the DJI MAVIC AIR case study file (android_logical.zip, DF048.E01 &DF048.E01 internal_microSD.0001).
- I obtain some basic information such as the MD5 & SHA256 hash values
- I start analyzing the DJI MAVIC AIR case study file (android_logical.zip, DF048.E01 &DF048.E01) using different digital forensic tools

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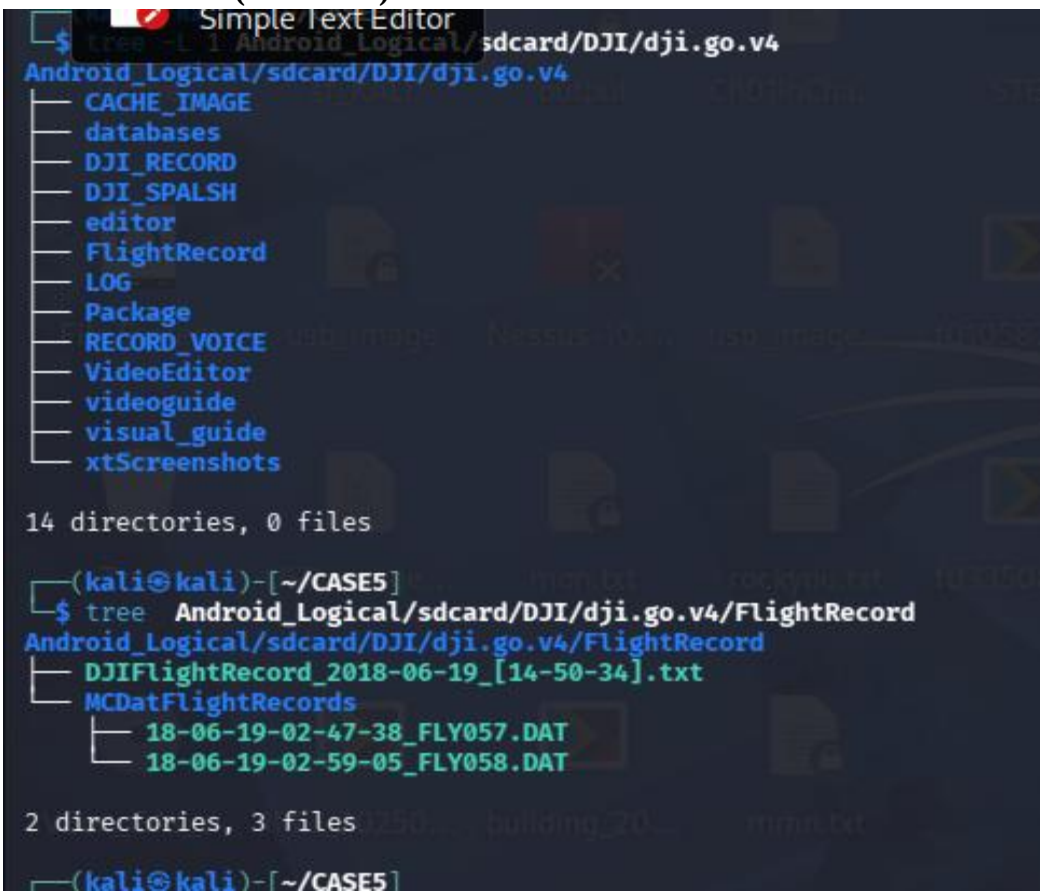
Screenshot and Evidence

EVIDENCES COLLECTED FROM THE DJI MAVIC AIR MOBILE CASE STUDY FILE(ANDRIOD LOGICAL)

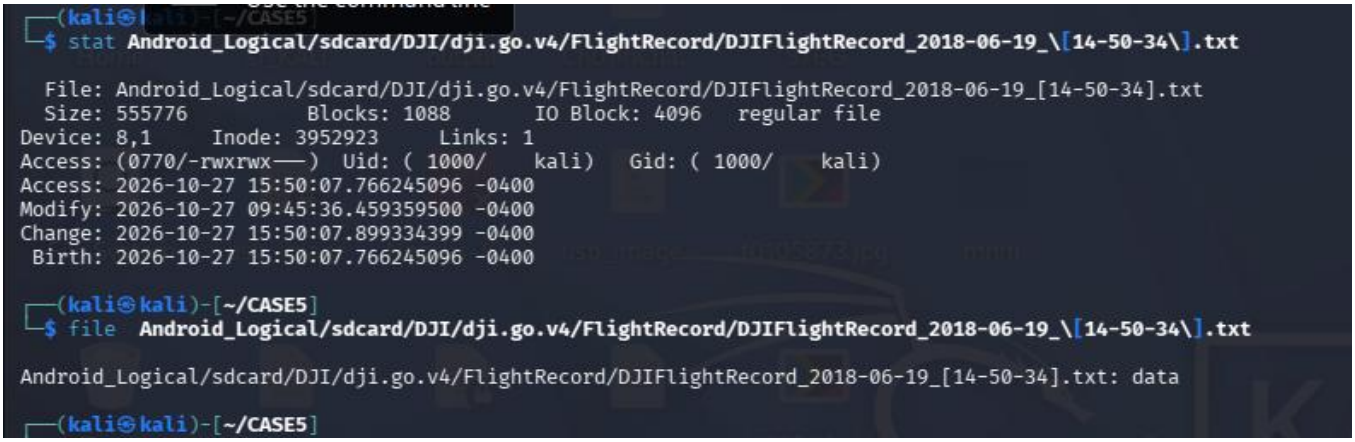
SCREENSHOT SHOWING THE HASH VALUE(MD5SUM) OF THE ANDROID LOCICAL FILE



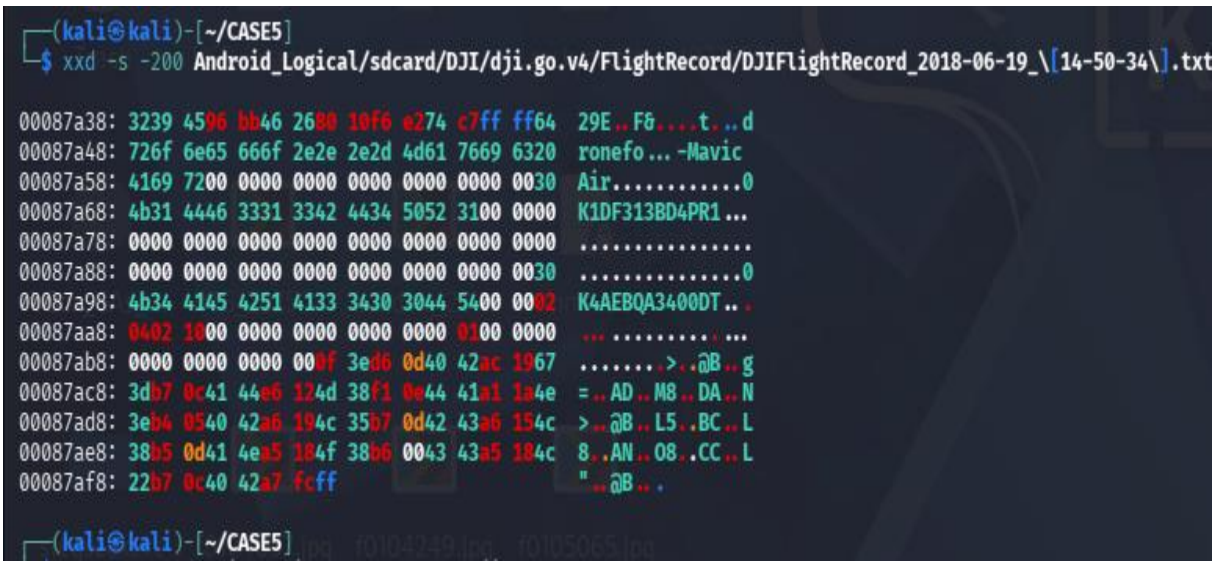
SCREENSHOT SHOWING THE FILE STRUCTURE OF THE DRONE CONTROL APPLICATION (DJI GO 4)



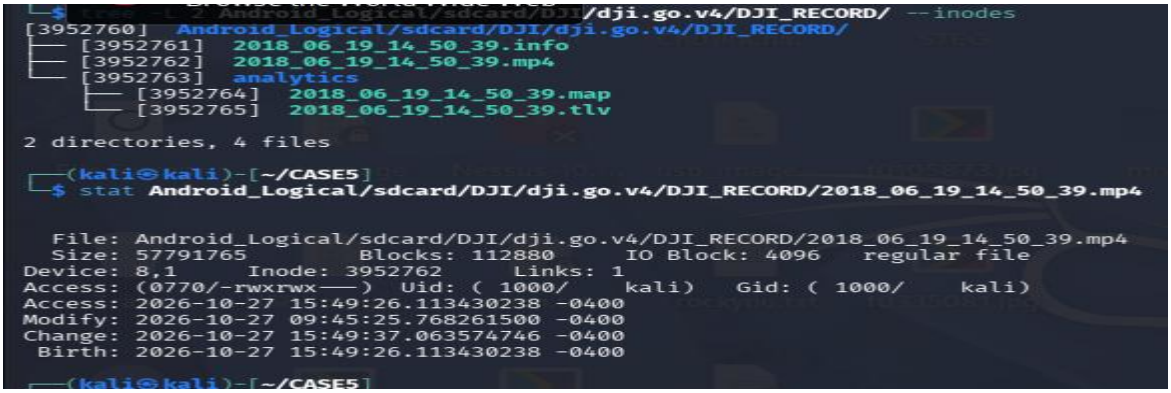
SCREENSHOT SHOWING SOME INFORMATION ABOUT THE FLIGHT RECORD



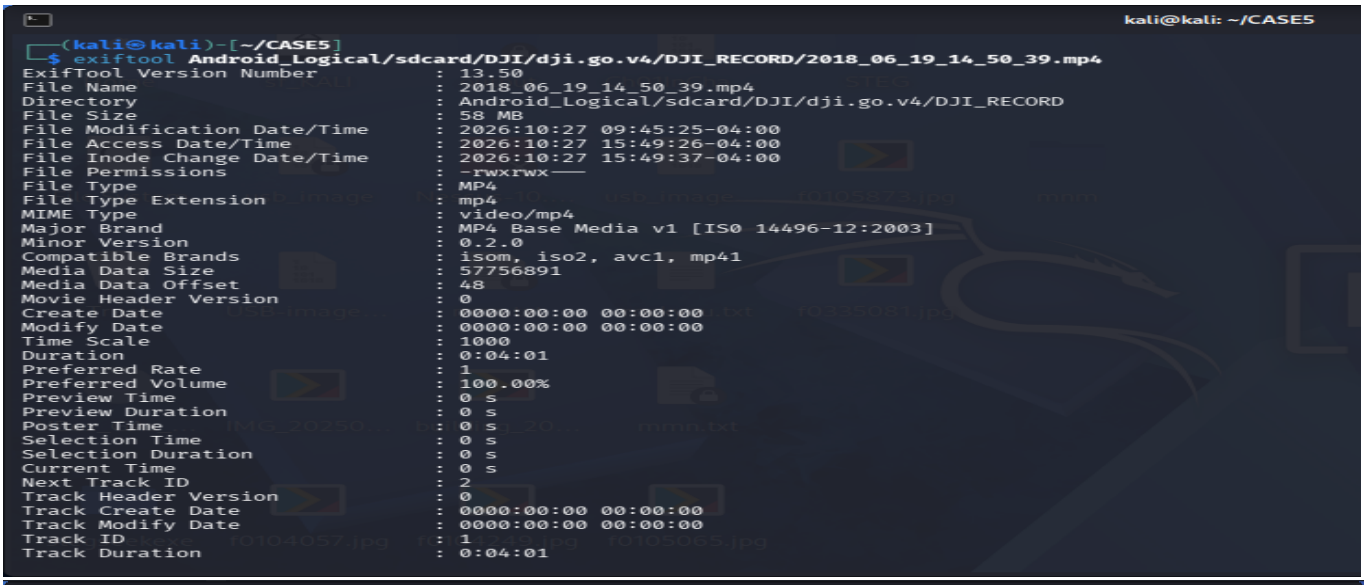
SCREENSHOT SHOWING EXTRACTION OF THE DEVICE NAME



SCREENSHOT SHOWING THE CAPUTURED RECORDING BY THE DRONE



SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT DRONE CAPTURED RECORDING

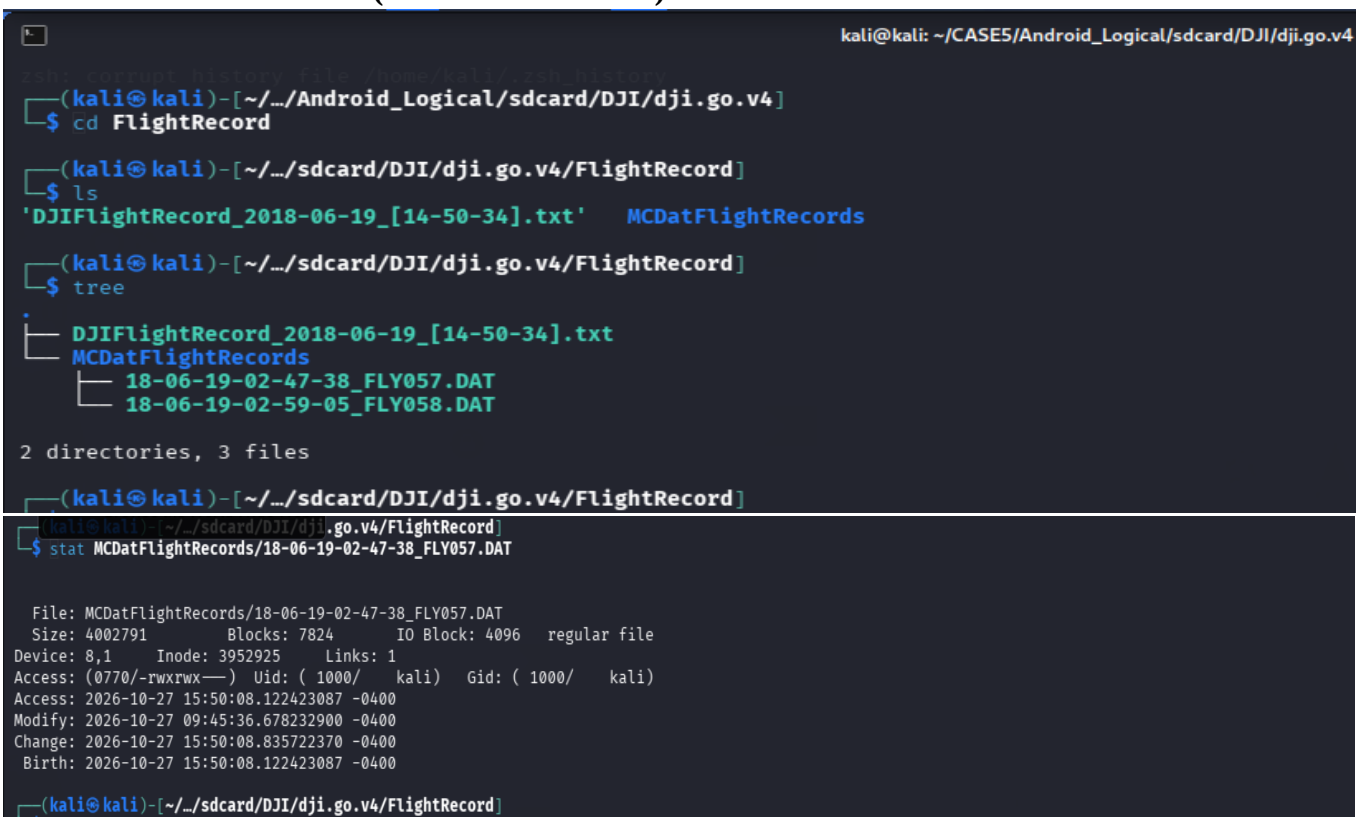




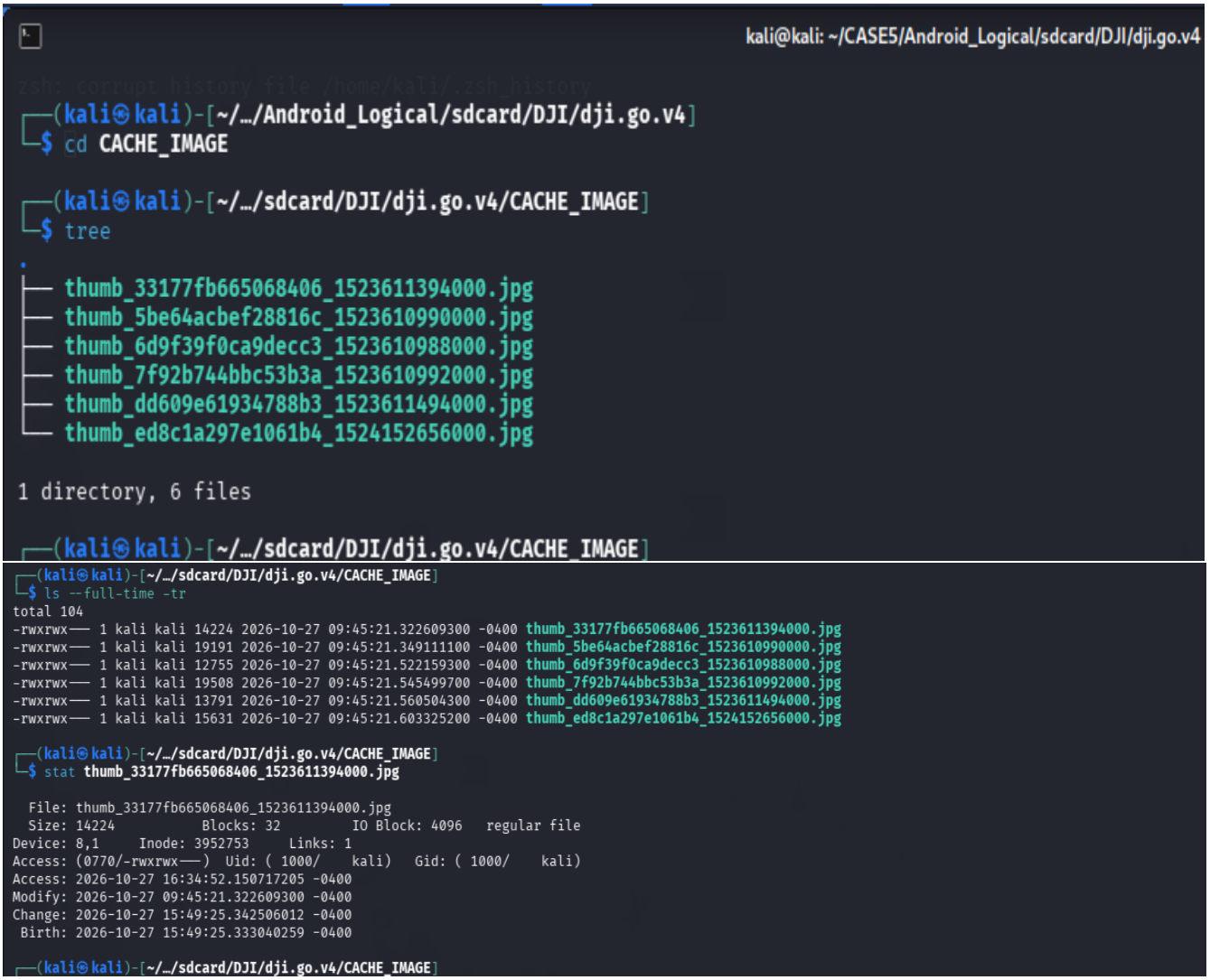
SCREENSHOT SHOWING THE DRONE CATURED RECORD



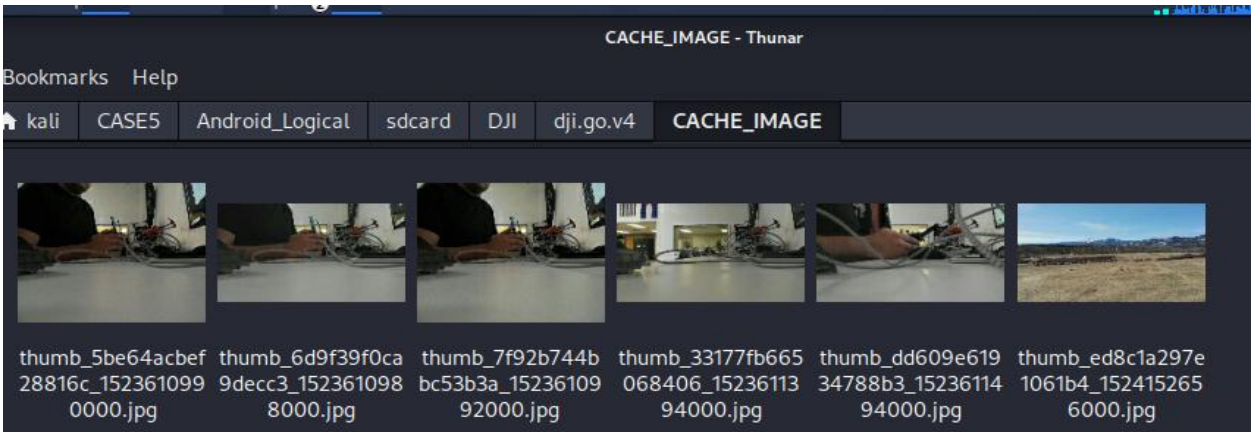
SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT THE DRONE FLYING INFORMATION(FLIGHT RECORD)



SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT THE CACHE IMAGE

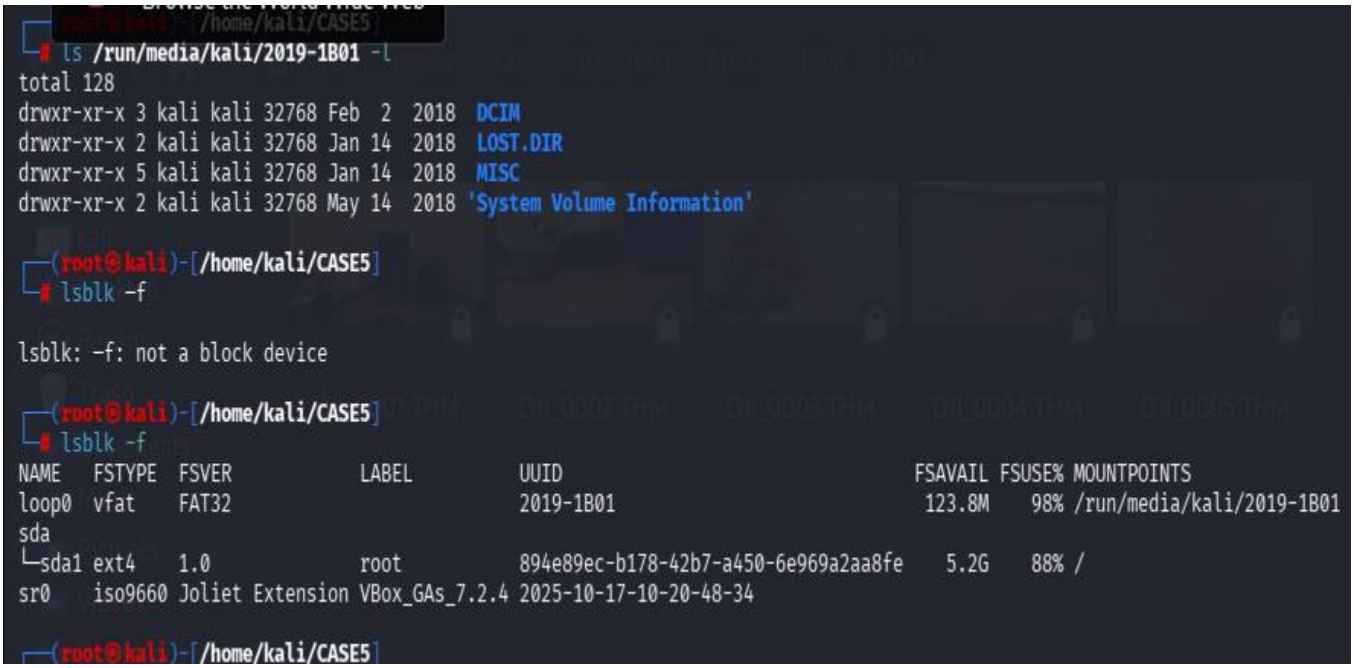


SCREENSHOT SHOWING ALL THE CACHE IMAGES

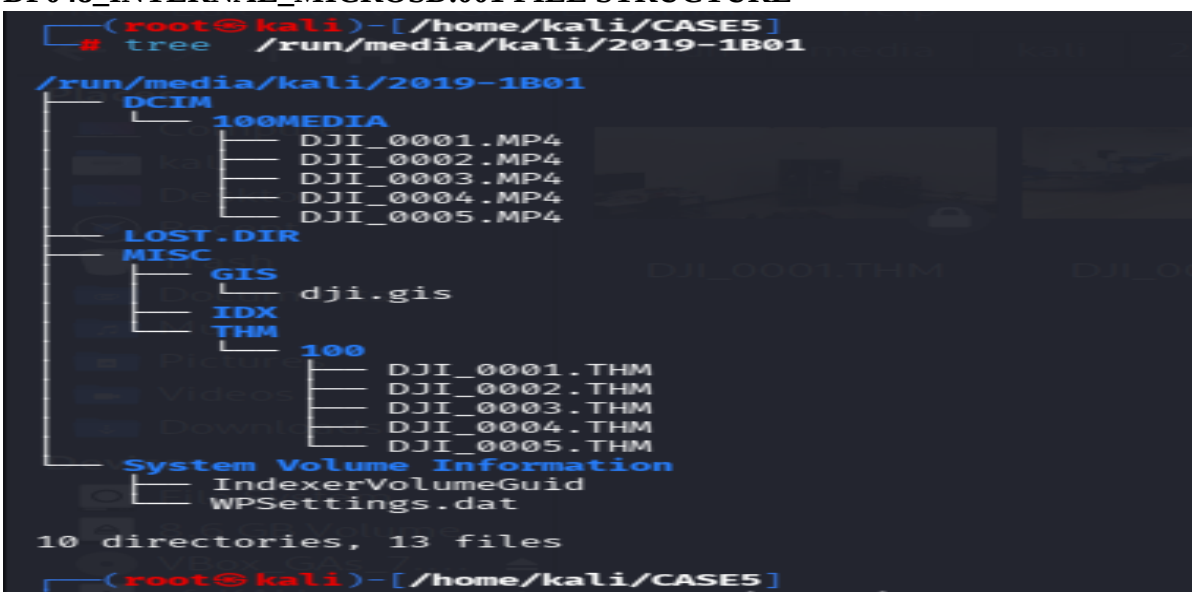


EVIDENCES COLLECTED FROM THE DJI MAVIC AIR MOBILE CASE FILE(DF048_INTERNAL_MICROSD.001)

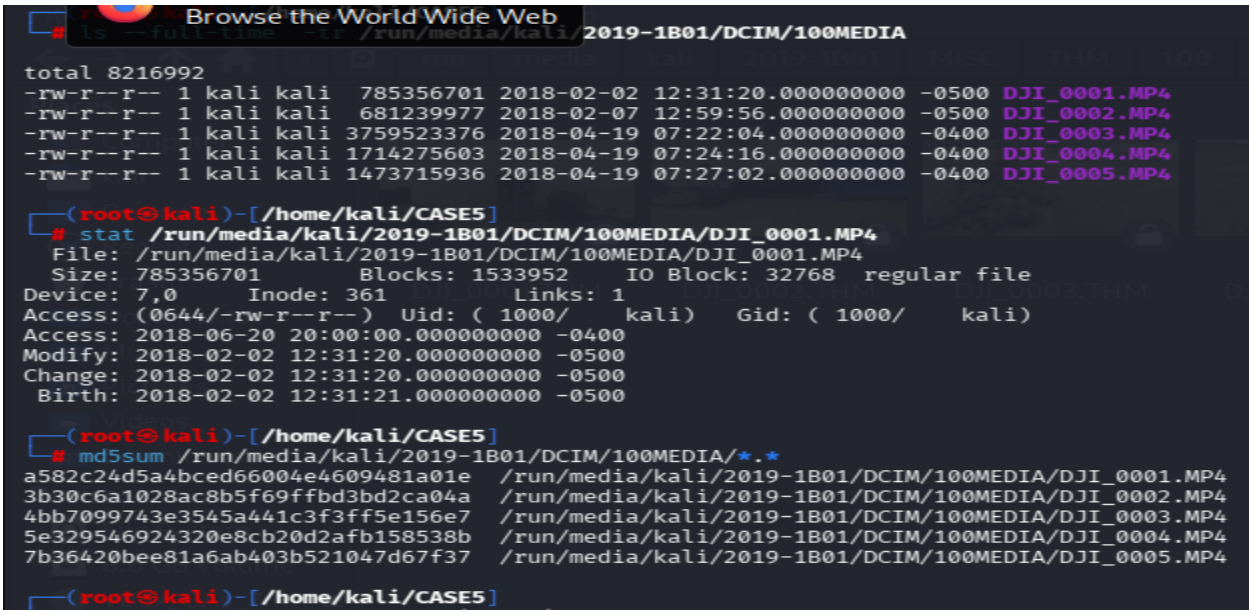
SCREENSHOT SHOWING SUCCESSFUL MOUNTING OF THE DF048_INTERNAL_MICROSD.001



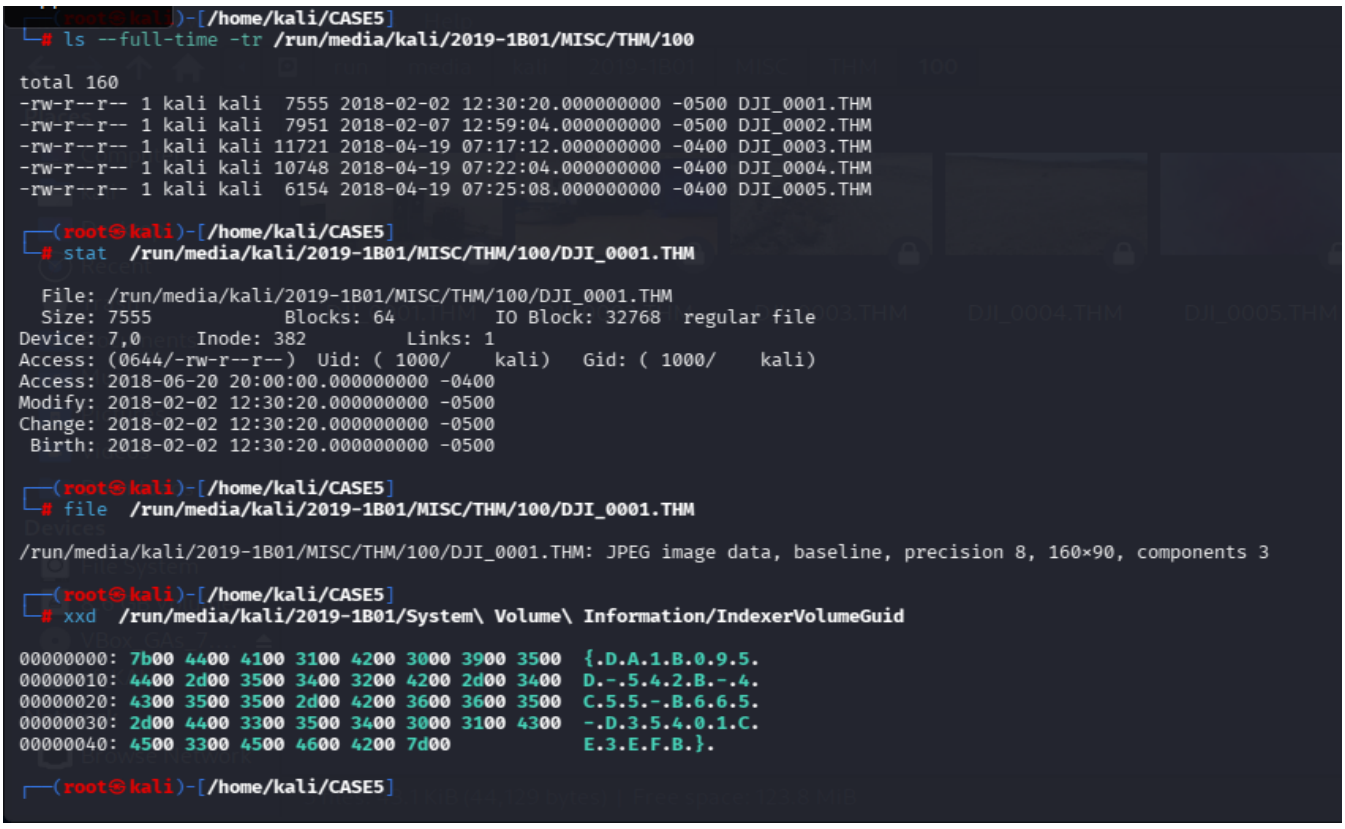
SCREENSHOT SHOWING A TREE DISPLAY OF THE DF048_INTERNAL_MICROSD.001 FILE STRUCTURE



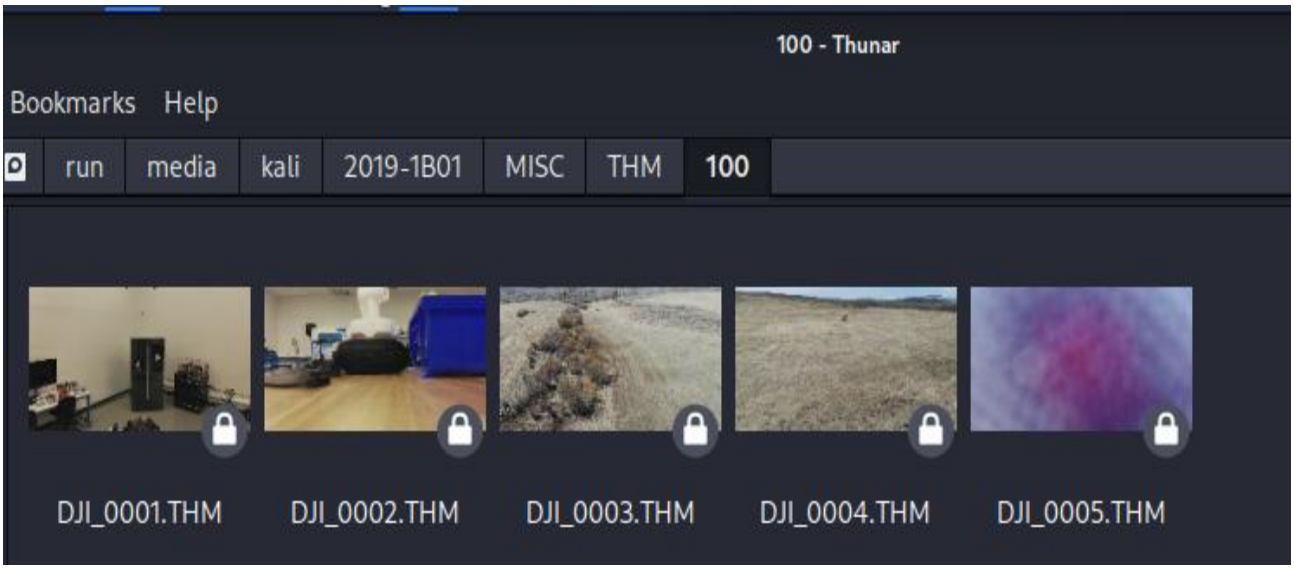
SCREENSHOT SHOWING THE LIST OF ALL THE MP4 FILE WITH THEIR MD5SUM HASH VALUE



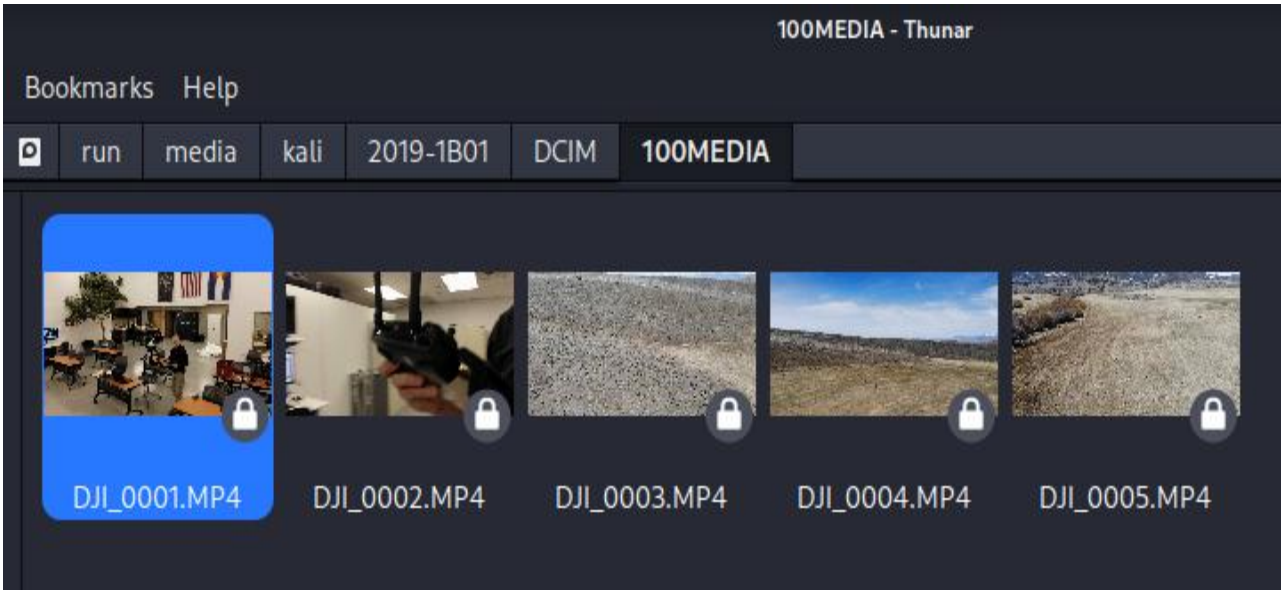
SCREENSHOT SHOWING A BASIC INFORMATION ABOUT A PARTICULAR THM FILE



SCREENSHOT SHOWING THE LIST OF ALL THE IMAGE FILE CONTAIN INT THE DF048_INTERNAL_MICROSD.001

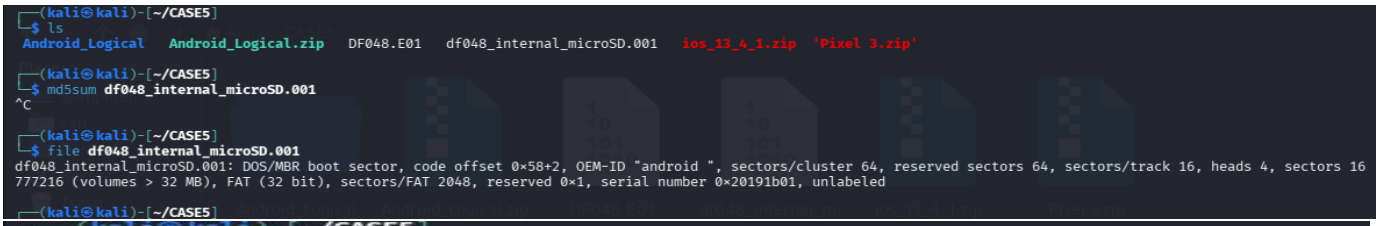


SCREENSHOT SHOWING THE LIST OF ALL THE VIDEO FILE CONTAIN INT THE DF048_INTERNAL_MICROSD.001



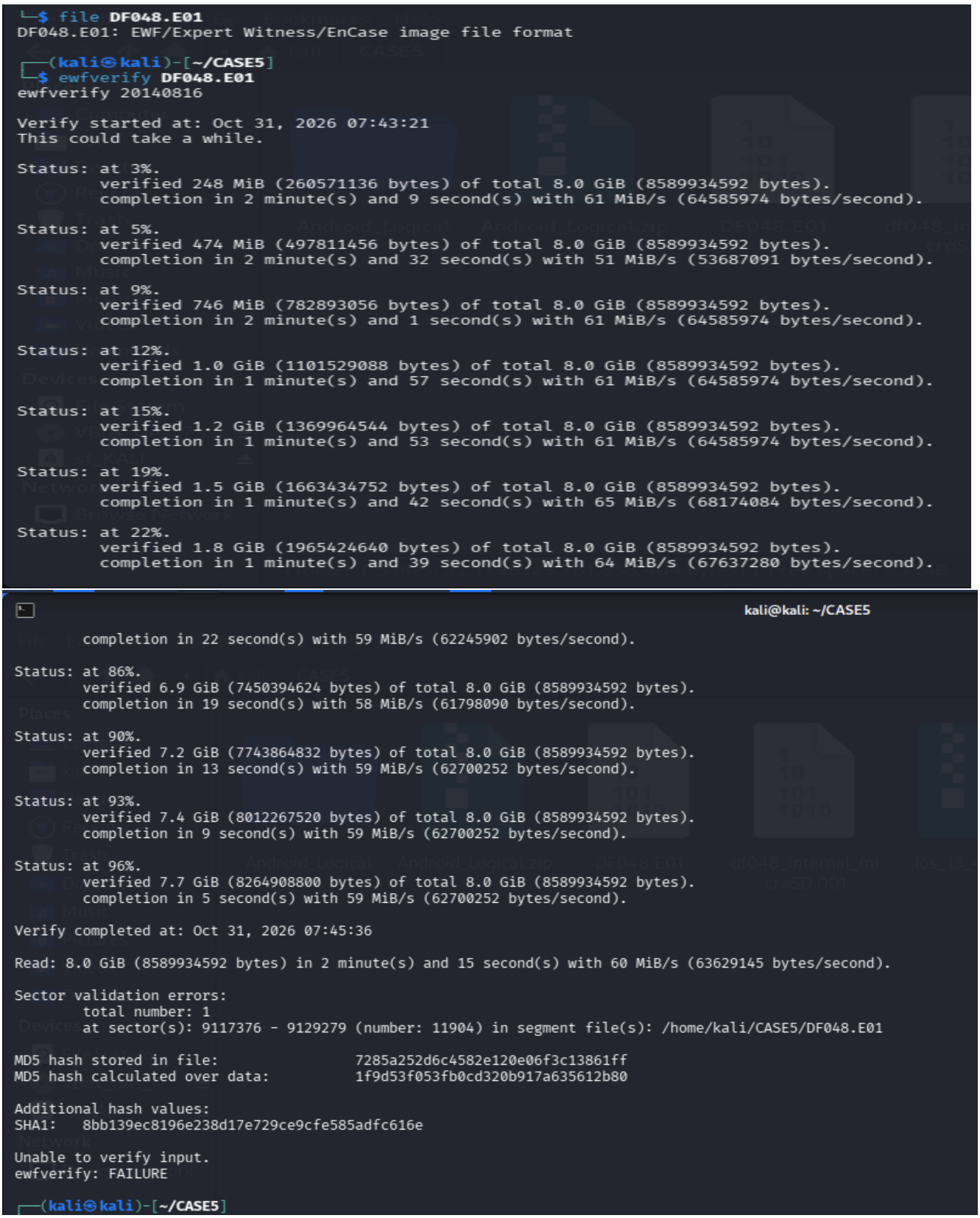
EVIDENCES COLLECTED FROM THE DJI MAVIC AIR MOBILE CASE FILE(DF048.E01)

SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT THE DF048.E01 FILE





SCREENSHOT SHOWING VERIFICATION OF THE DF438.E01 HASH VALUE



```
└─$ file DF048.E01
DF048.E01: EWF/Expert Witness/EnCase image file format

┌──(kali㉿kali)-[~/CASE5]
└─$ ewfverify DF048.E01
ewfverify 20140816

Verify started at: Oct 31, 2026 07:43:21
This could take a while.

Status: at 3%.
        verified 248 MiB (260571136 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 2 minute(s) and 9 second(s) with 61 MiB/s (64585974 bytes/second).

Status: at 5%.
        verified 474 MiB (497811456 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 2 minute(s) and 32 second(s) with 51 MiB/s (53687091 bytes/second).

Status: at 9%.
        verified 746 MiB (782893056 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 2 minute(s) and 1 second(s) with 61 MiB/s (64585974 bytes/second).

Status: at 12%.
        verified 1.0 GiB (1101529088 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 1 minute(s) and 57 second(s) with 61 MiB/s (64585974 bytes/second).

Status: at 15%.
        verified 1.2 GiB (1369964544 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 1 minute(s) and 53 second(s) with 61 MiB/s (64585974 bytes/second).

Status: at 19%.
        verified 1.5 GiB (1663434752 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 1 minute(s) and 42 second(s) with 65 MiB/s (68174084 bytes/second).

Status: at 22%.
        verified 1.8 GiB (1965424640 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 1 minute(s) and 39 second(s) with 64 MiB/s (67637280 bytes/second).
```

```
kali@kali: ~/CASE5

        completion in 22 second(s) with 59 MiB/s (62245902 bytes/second).

Status: at 86%.
        verified 6.9 GiB (7450394624 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 19 second(s) with 58 MiB/s (61798090 bytes/second).

Status: at 90%.
        verified 7.2 GiB (7743864832 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 13 second(s) with 59 MiB/s (62700252 bytes/second).

Status: at 93%.
        verified 7.4 GiB (8012267520 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 9 second(s) with 59 MiB/s (62700252 bytes/second).

Status: at 96%.
        verified 7.7 GiB (8264908800 bytes) of total 8.0 GiB (8589934592 bytes).
        completion in 5 second(s) with 59 MiB/s (62700252 bytes/second).

Verify completed at: Oct 31, 2026 07:45:36

Read: 8.0 GiB (8589934592 bytes) in 2 minute(s) and 15 second(s) with 60 MiB/s (63629145 bytes/second).

Sector validation errors:
        total number: 1
        at sector(s): 9117376 - 9129279 (number: 11904) in segment file(s): /home/kali/CASE5/DF048.E01

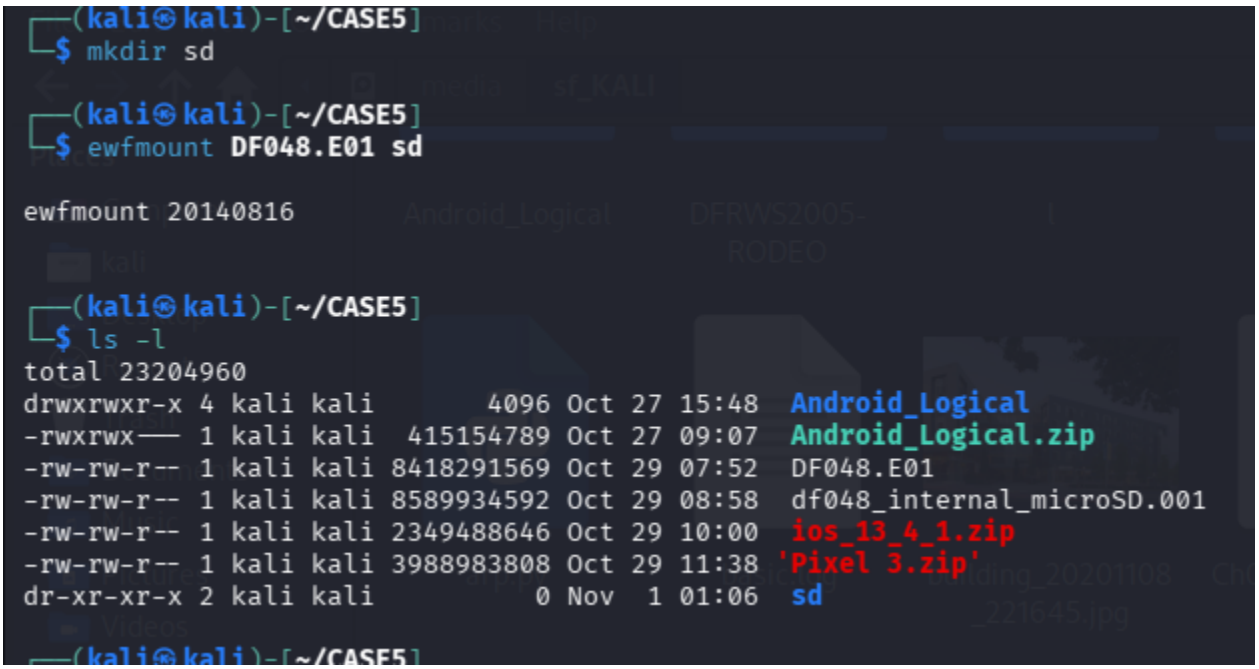
MD5 hash stored in file:                7285a252d6c4582e120e06f3c13861ff
MD5 hash calculated over data:          1f9d53f053fb0cd320b917a635612b80

Additional hash values:
SHA1:   8bb139ec8196e238d17e729ce9cfe585adfc616e

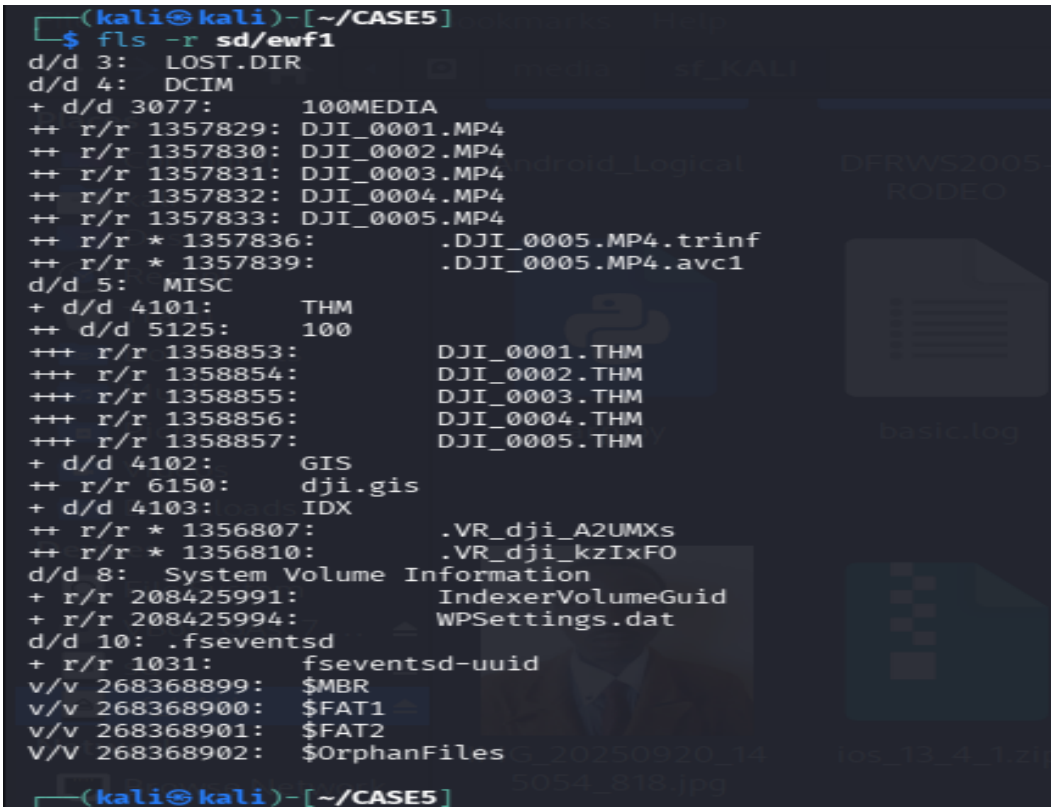
Unable to verify input.
ewfverify: FAILURE

┌──(kali㉿kali)-[~/CASE5]
```

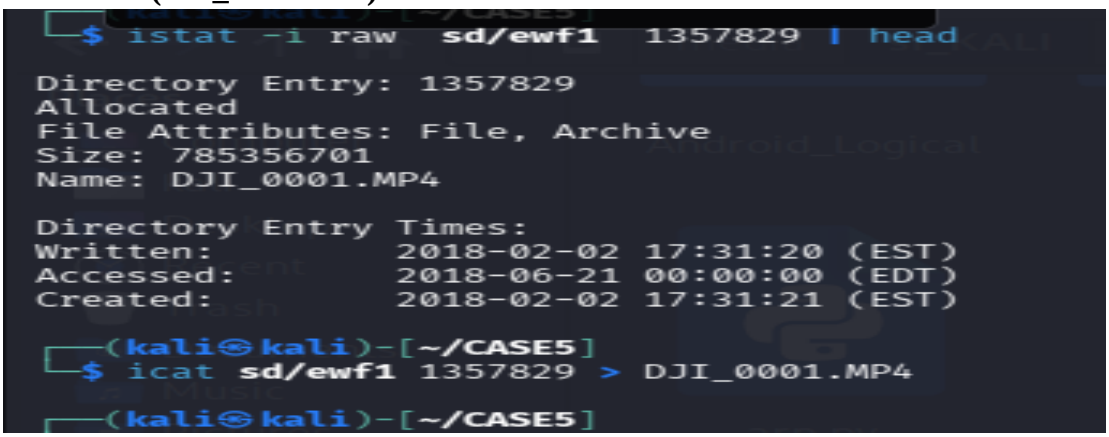
SCREENSHOT SHOWING MOUNTING OF THE DF048.E01 FILE



SCREENSHOT SHOWING THE LIST OF FILE INSIDE THE DF048.E01 FILE



SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT A PARTICULAR MP4 FILE(DJI_0001.MP4)



SCREENSHOT SHOWING THE HASH VALUE OF THE MP4 FILE(DJI_0001.MP4)



SCREENSHOT SHOWING A DISPLAY OF THE MP4 FILE(DJI_0001.MP4)



Analysis and Discussion

With the above screenshot/evidences I have collecterd above, I can conclude that the device name is Mavic Air and also the MP4 file available inside the DJI GO 4 application is different from one inside the two microSD. While the DF048_INTERNAL_MICROSD.001 and DF048.E01are both the same because they both contain the same MP4 file and THM file .

Conclusion

In conclusion these lab has provide me with the basic skill I will required when analyzing/handling a digital evidence related to a drone system.

Reference

➢ICDFA course Material (SBT-DF204(computer forensic case studies) by COMMANDANT AMINU IDRIS